



Project 8 — AI-based Patent Analysis, Design-Around & Innovation Recommendation

Partner	Fuyao Europe
Focus	Industrial and Applied Application
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Sync	Monthly
Resources	Datasets

Project Details

The goal of this project is to develop an AI-supported system that can read and understand patent documents, compare them with product designs, identify potential risks, and provide suggestions for design improvement and innovation. The project is structured in four phases with accompanying evaluation criteria.

Phase 1: Patent Understanding and Data Structuring

Collect and review patent documents (e.g., claims and descriptions). Extract key technical features and organize them in a structured way. Build a simple database or schematic diagram for further analysis.

Phase 2: Risk Identification

Compare product design inputs (e.g., key features or simplified descriptions) with patent content. Identify overlapping areas and potential risks. Highlight critical features that may be sensitive.

Phase 3: Design Improvement Suggestions

Propose alternative design ideas to reduce or avoid risks. Ensure suggestions are realistic and feasible from an engineering perspective. Provide clear and structured outputs.

Phase 4: Innovation Opportunities

Review multiple patents to identify common patterns and gaps. Suggest potential directions for new ideas or improvements.

Evaluation Criteria / KPIs

- **Accuracy of Information Extraction** — How correctly key features are identified from patents.
- **Risk Identification Quality** — How well potential conflicts are detected.
- **Quality of Suggestions** — Usefulness and feasibility of design improvement ideas, evaluated by experts.
- **Innovation Relevance** — Whether proposed ideas are meaningful and practical.
- **Usability** — Clarity and usefulness of the system from a user perspective.

About the Partner

Fuyao Europe, headquartered in Leingarten near Heilbronn, is the German subsidiary of the China Fuyao Glass Industry Group — the world's largest provider of integrated automotive glass and trim solutions. Customers include JLR, Volkswagen Group, GM, BMW, Mercedes-Benz, Ford, Stellantis, and others.